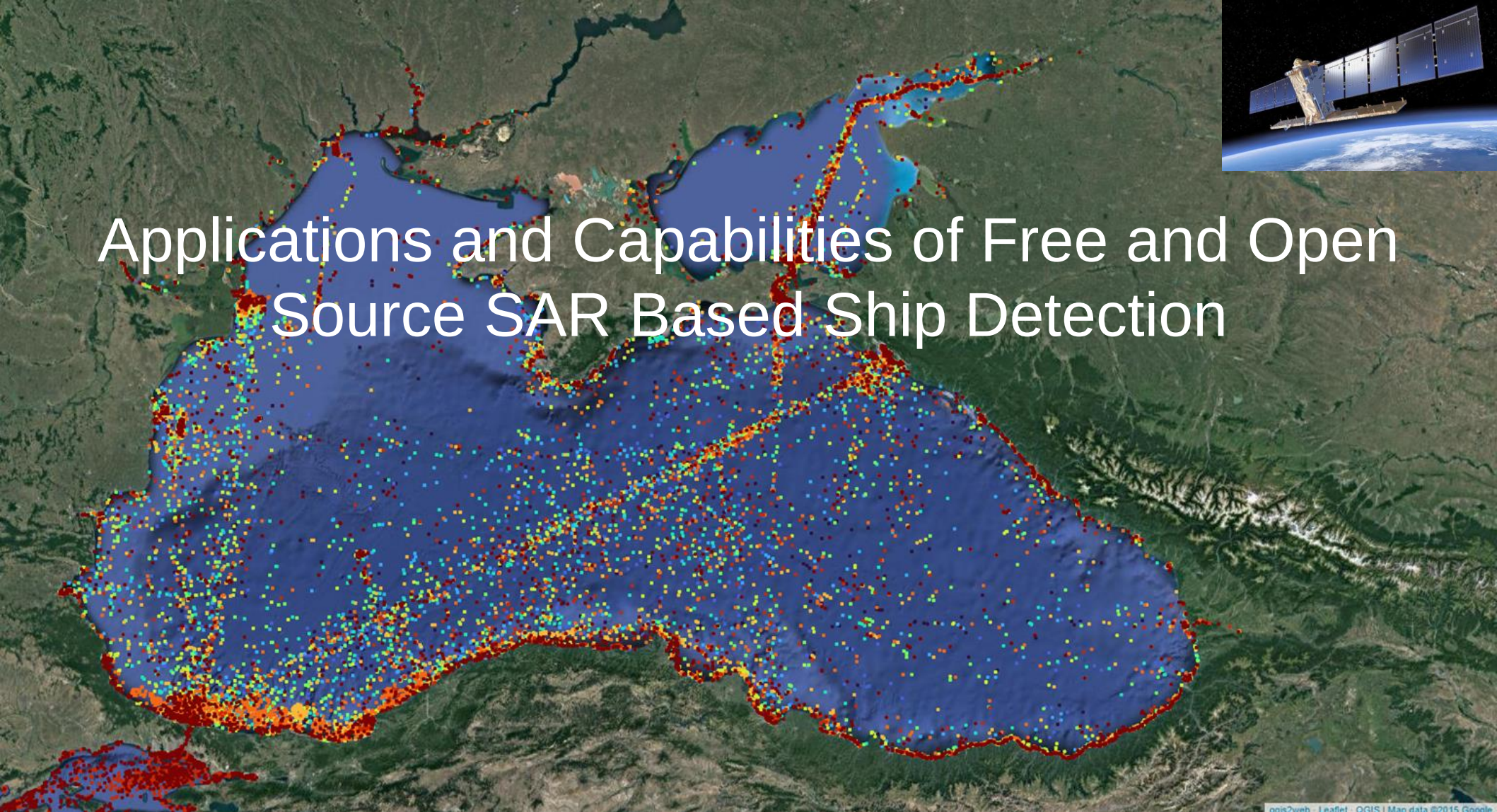


Applications and Capabilities of Free and Open Source SAR Based Ship Detection



Overview

- Motivation
- Methodology
- Results and Applications
- Next Steps

Motivation

- All current ship tracking information is based on voluntarily broadcast AIS signals
- Issues with locating ships without transmitting transponders
- Illegal fishing
- Military operations
- Piracy
- Human trafficking

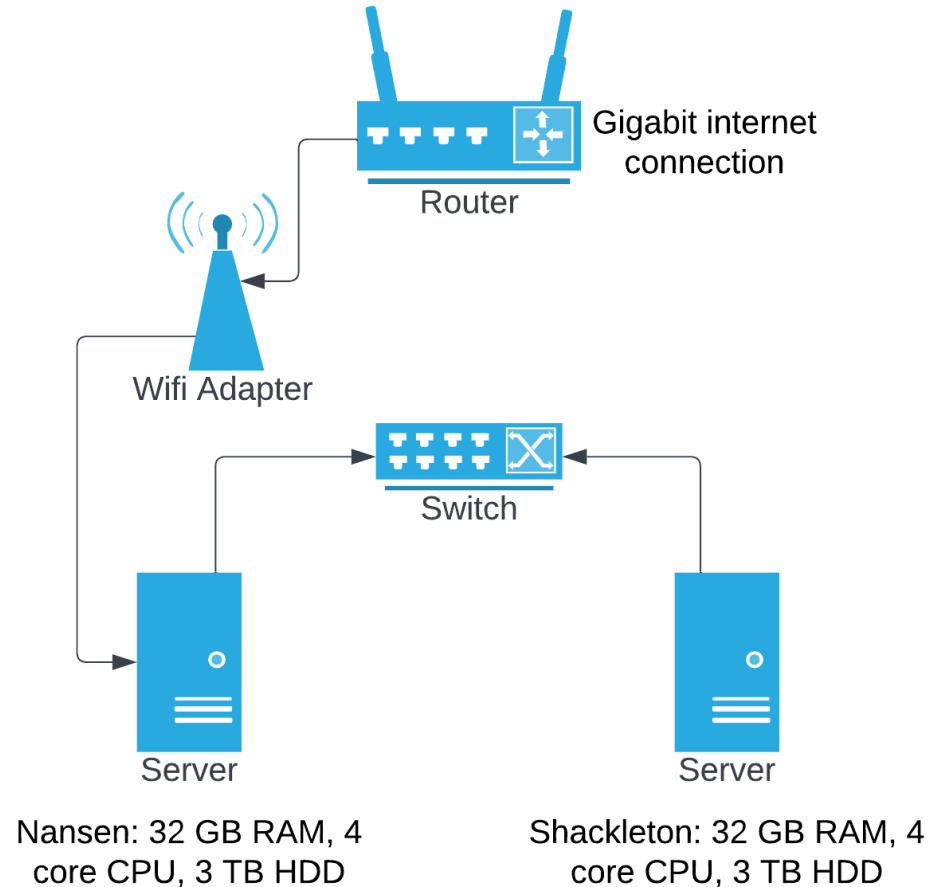


Current Applications

- Most applications in the motivation would require the integration and correlation of AIS data which is not currently implemented. Expected applications in its current state include:
 - Monitoring traffic in shipping lanes, port backlogs and other bottlenecks
 - Observing fisheries
- Unanticipated applications include:
 - Monitoring offshore wind farm construction
 - Offshore oil operations

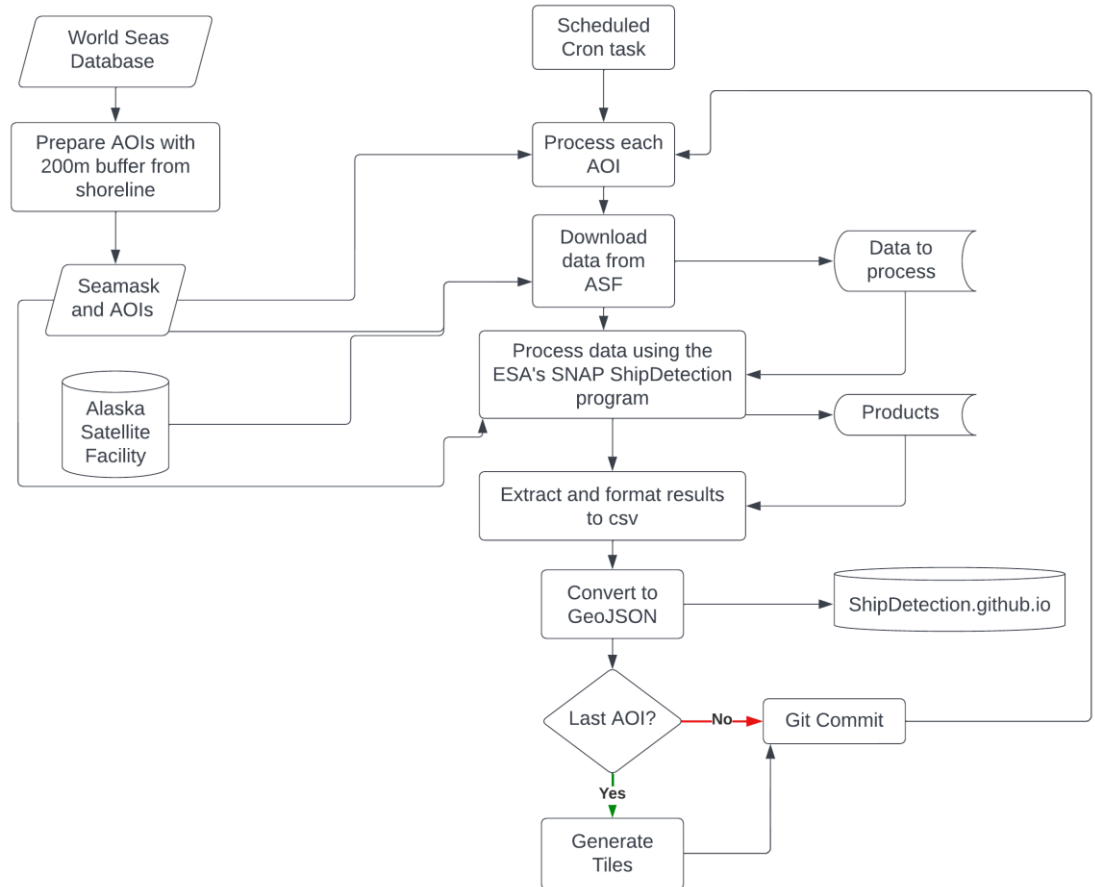
Processing Resources

- All processing occurs using 2 Lenovo Thinkcentres
- Processing AOIs shared between them and results output over the network to a single github repository to reduce risk of conflicts.
- Each with:
 - 32 GB RAM
 - 3TB HDD Storage
 - 4 core I5 CPU
 - Gigabit internet over a shared wifi adapter through a switch
- Currently takes ~8hrs a day for the processing to complete. Highly dependent on internet speed.



Processing Workflow

- Download data from the ASF
- Run Shipdetection algorithm using ESA snap with a landmask with a 200 m buffer
- Extract the results from output directory
- Convert results into leaflet compatible json
- Generate vector tiles
- Commit to webpage

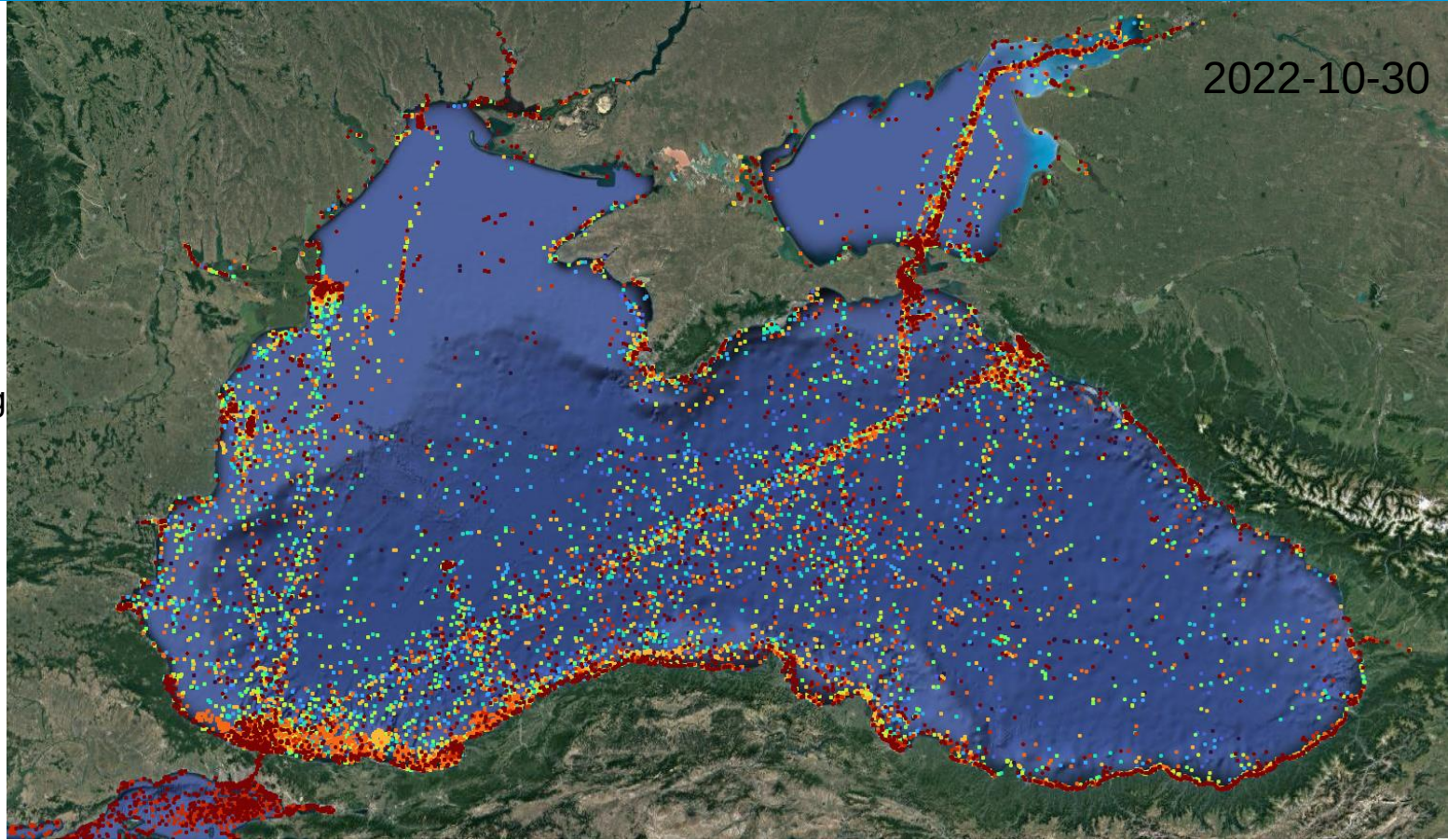


Results – Black Sea

Colour scale based on age of the detection (Red to blue with 12 day increments up to 120 days. Based on up to 10 revisits of SNT

Notable observations:

- Reduction in shipping backlog around the Danube Delta originally caused due to the blockade of Odesa
- Reopening of shipping lanes to the port of Odesa following the grain deal



Results - Mediterranean Sea

Manually digitized the Suez canal
to observe ships in transit

2022-11-13

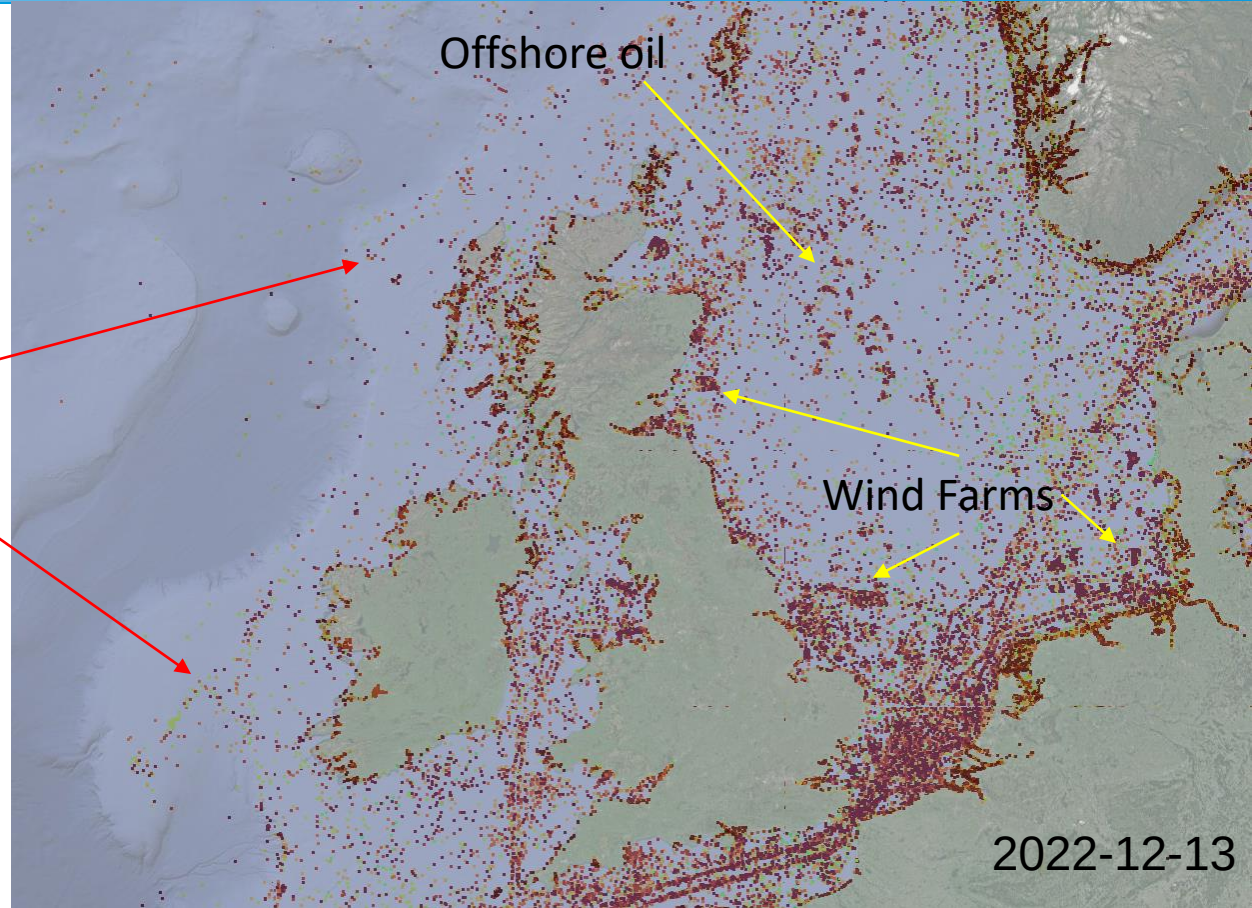
- Split into eastern and western med
- Can clearly see the shipping lanes from the Suez Canal, Turkish Straights and the Adriatic Sea.
- Something weird going on off the coast of Tunisia. Fishing?

3/25/2023

Results – North Sea

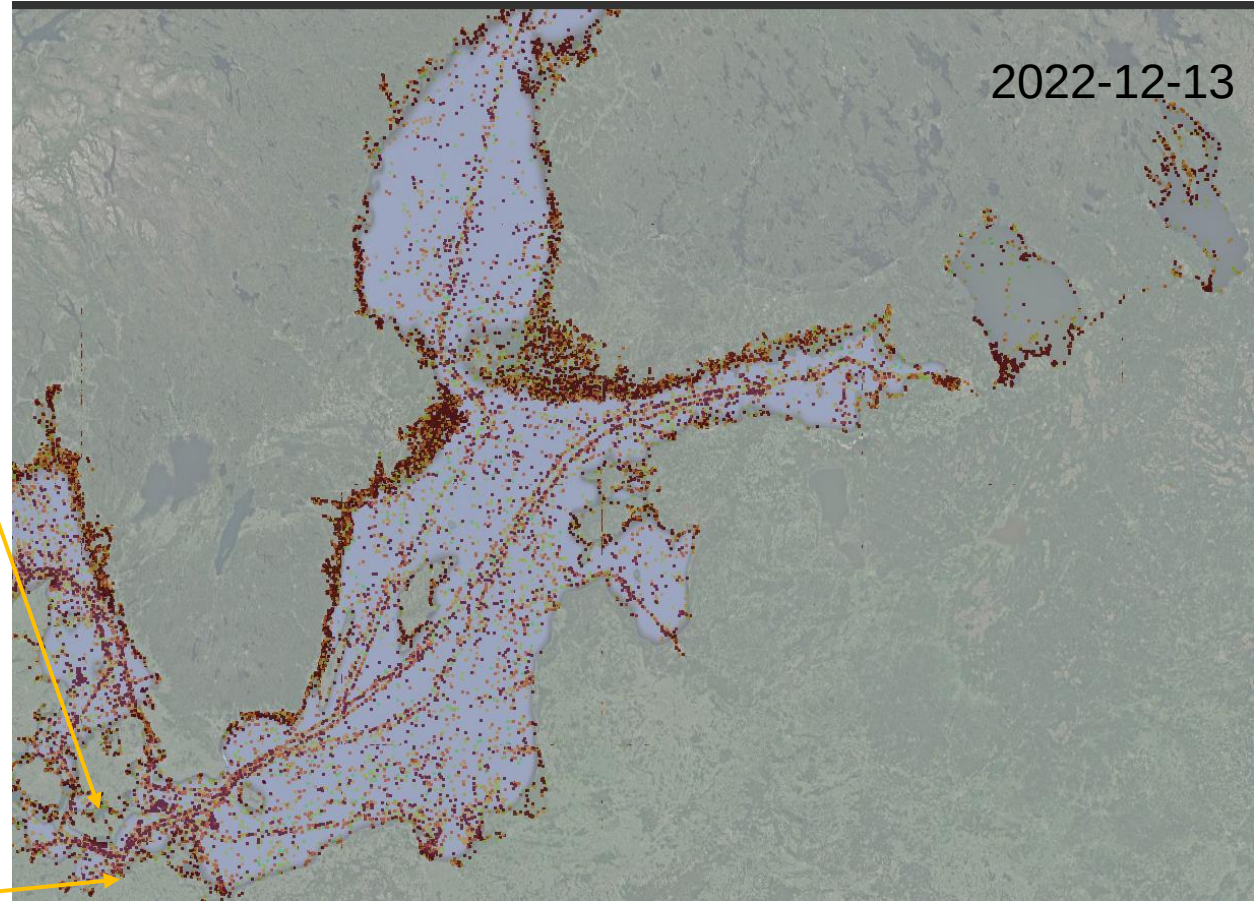
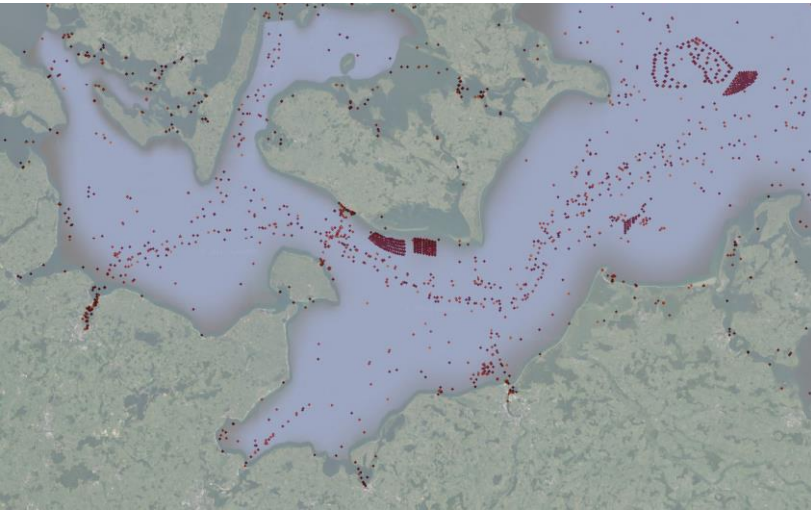
- Wind farms show up very well
- North Sea oil production highlighted
- Probable Location of fisheries indicated

Probable
Fisheries



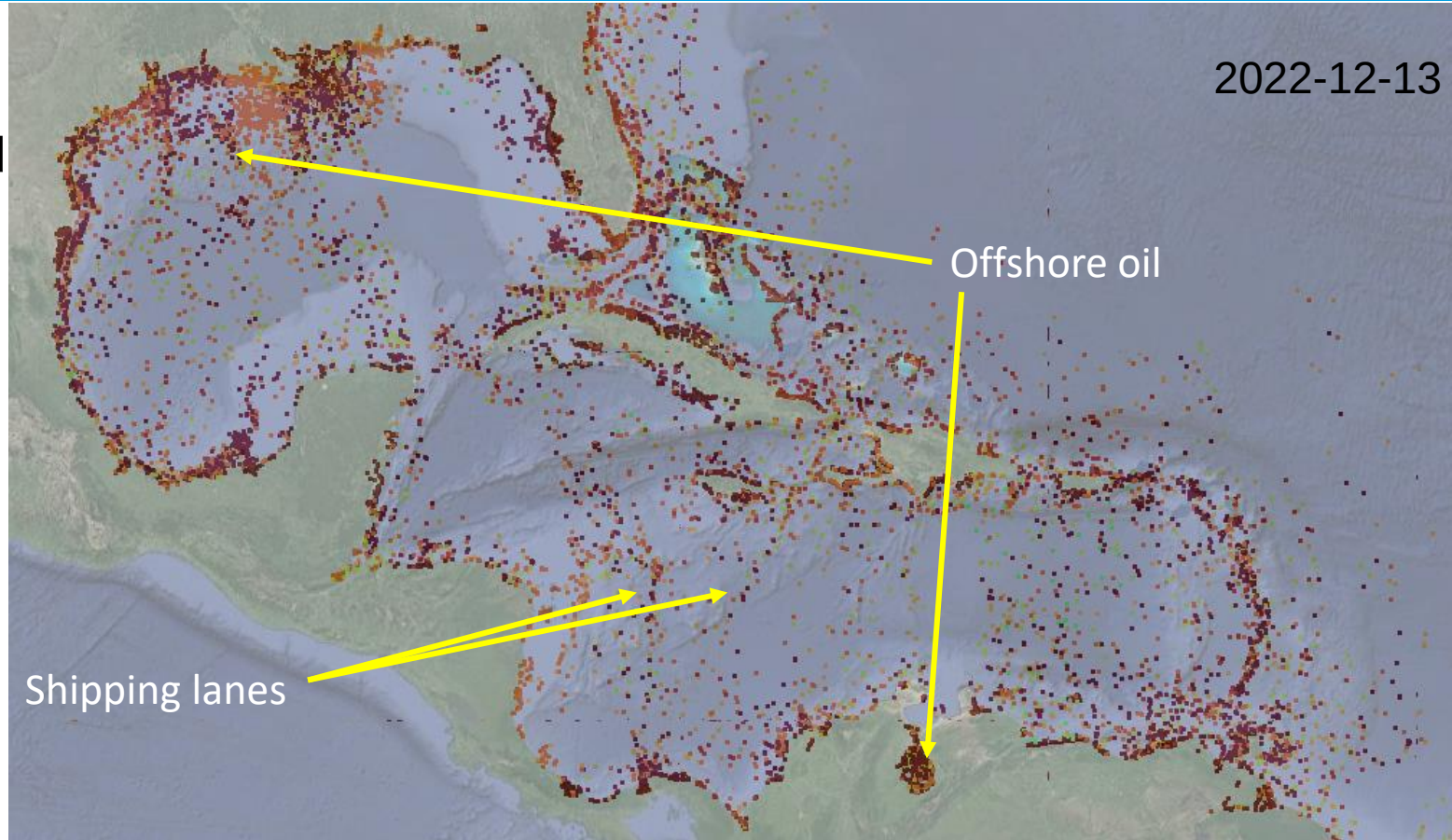
Results – Baltic Sea

- Shows main shipping lanes
- Ferry routes and wind farms are apparent
- Lot of false positives off the coast of Sweden and Finland due to small rocky islands not included in the seamask



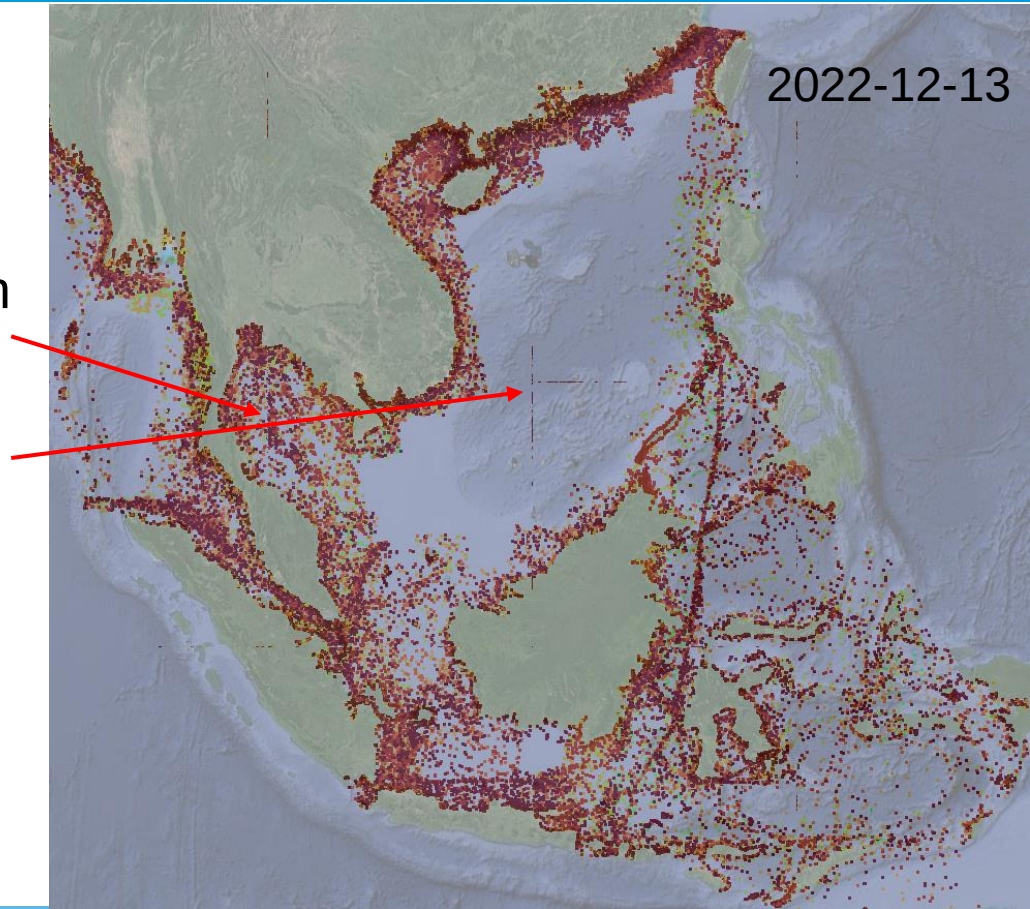
Results - Caribbean

- Offshore oil operations highlighted
- Shipping lanes to the Panama canal



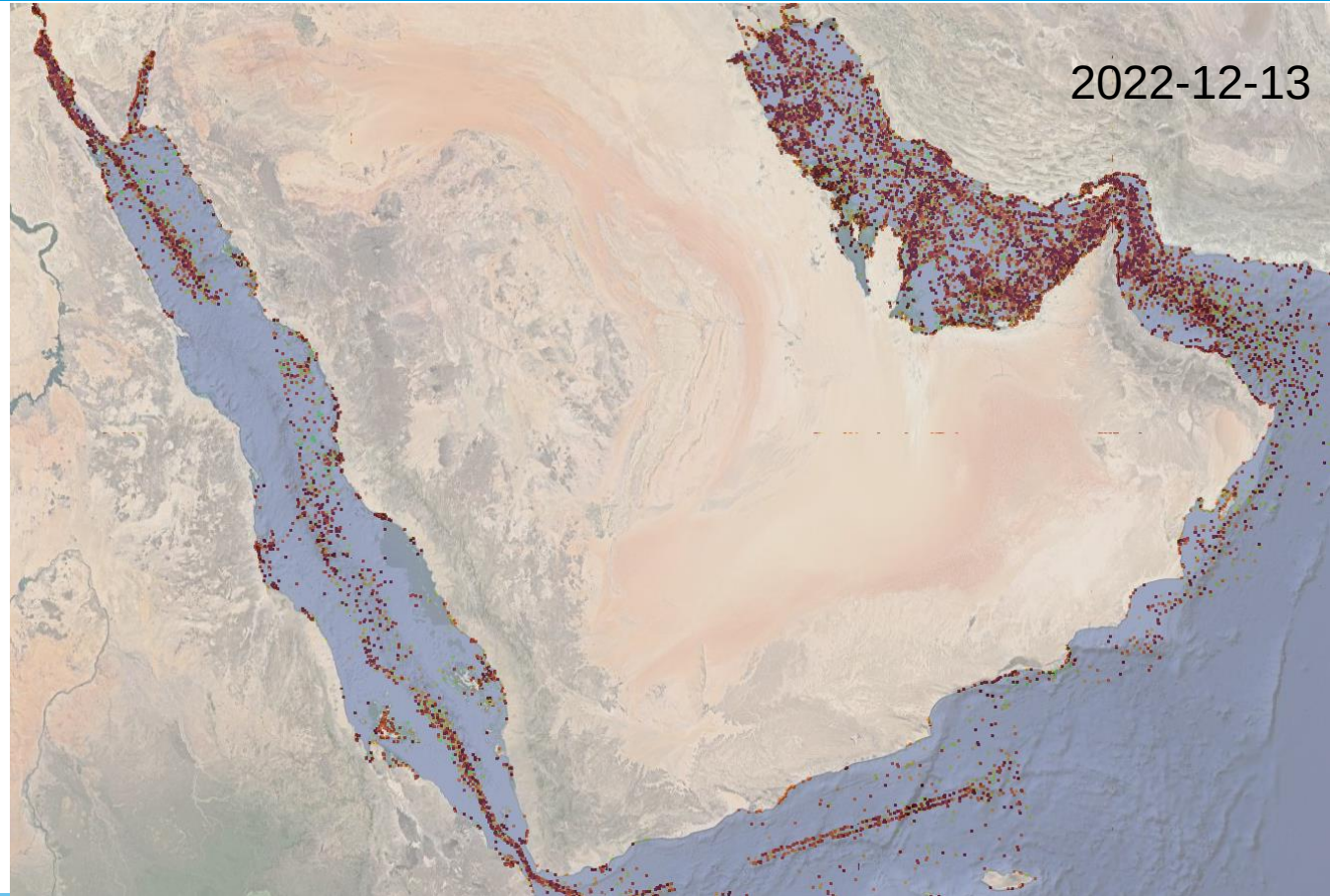
Results – Southeast Asia

- Main Shipping lanes are highlighted.
- Offshore oil in the Gulf of Thailand
- SNT is not acquired over a most of the South China Sea
- An issue is apparent here with edge effects on the borders of the tiles



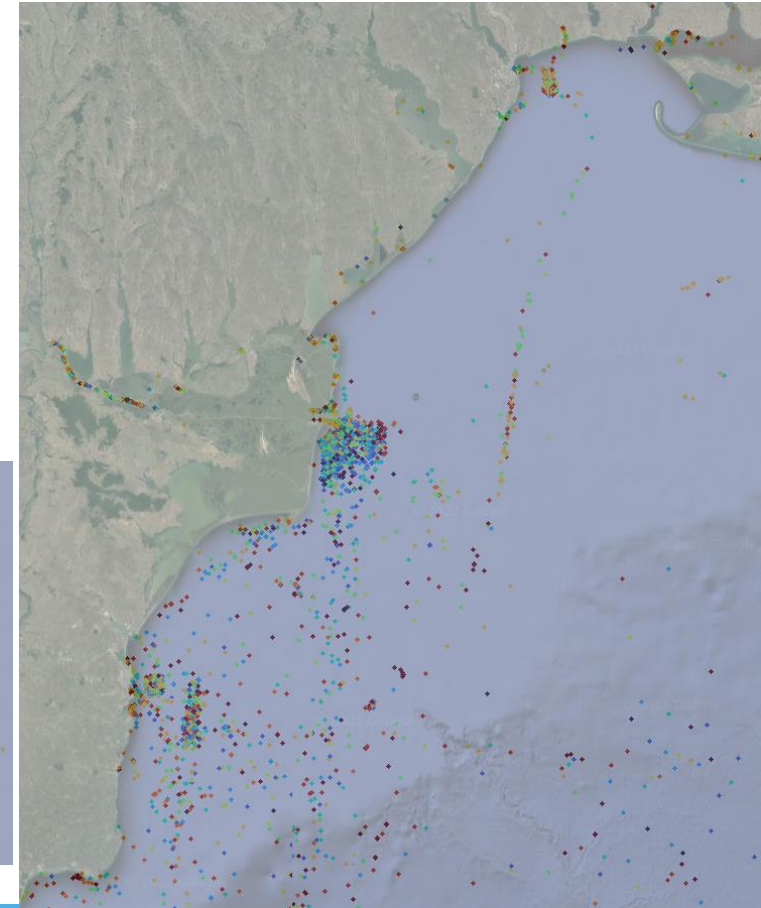
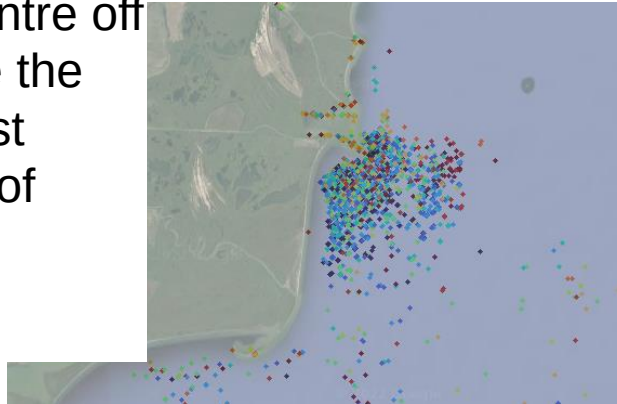
Results – Arabian Peninsula

- SNT does not acquire over a portion of the Red Sea
- Shipping lanes from the Suez canal highlighted.
- Offshore oil in the Persian Gulf highlighted



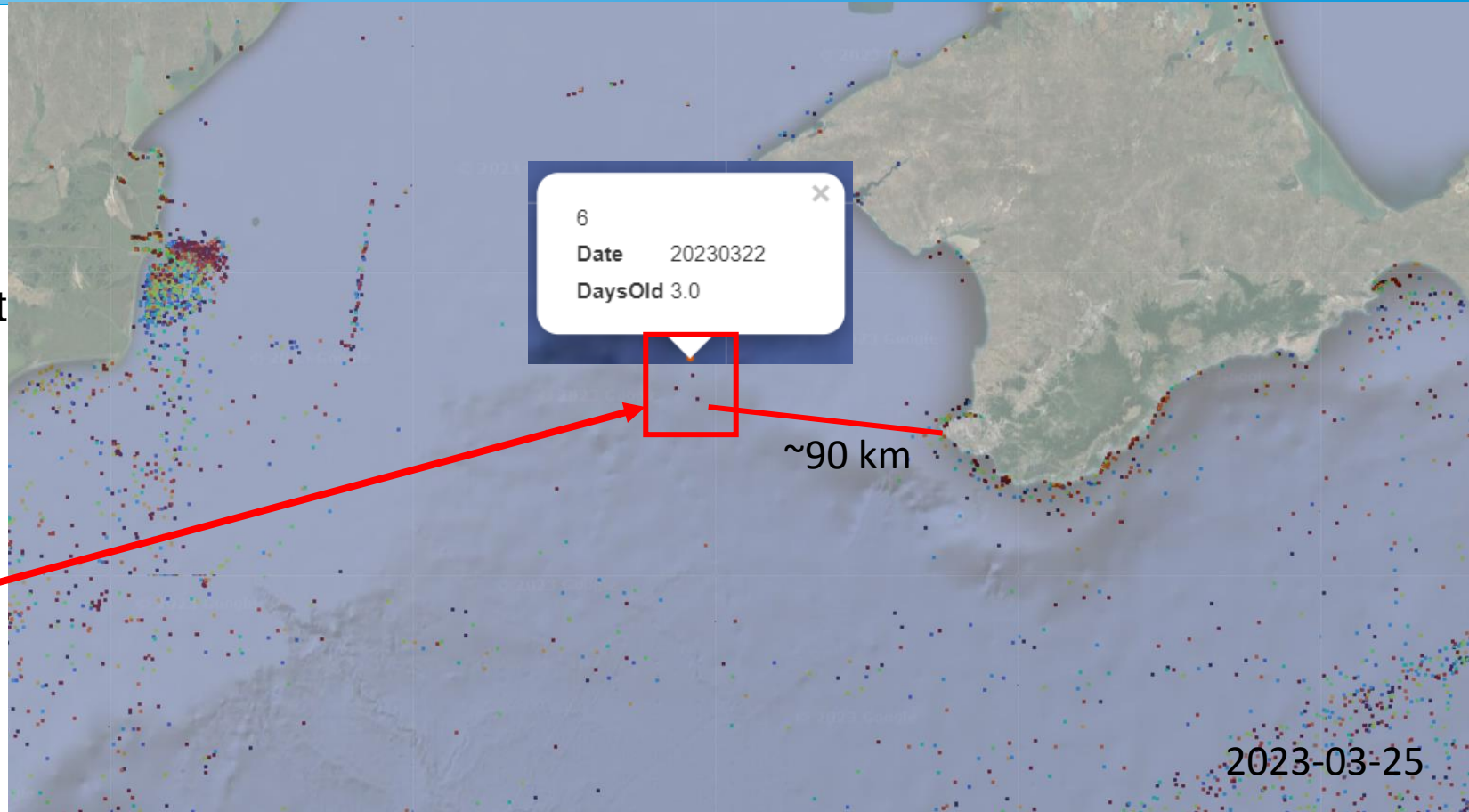
Applications – Changes to shipping routes

- Highlights changes to shipping activity
 - Good at indicating sudden backlogs, clearing of backlogs, increases and decreases in shipping lane usage
- Large area showing blue ships with a cluster of red ships in the centre off the coast of Romania, indicate the clearing of a backlog which first began with Russian blockage of Odesa



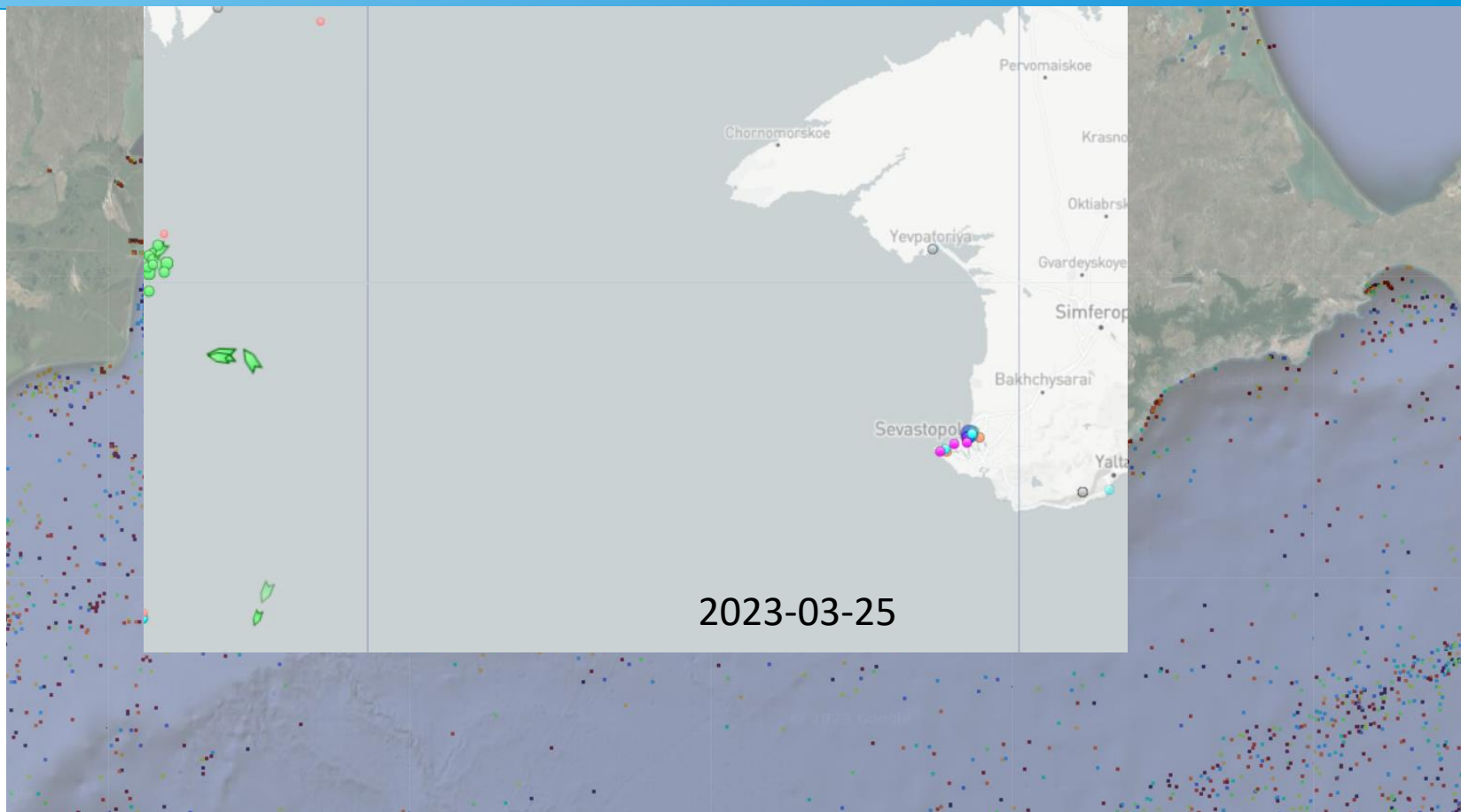
Applications – Naval Intelligence

- Three objects located near the suspected crash site of the American MQ-9 Reaper drone lost over the Black Sea on March 14th, 2023.
- No history of ships in this area since the beginning of the war.



Applications – Naval Intelligence

- No AIS transmissions from the area on marinetraffic
- One point is a repeat detection. Probably not a ship
- Likely two ships of the Russian black sea fleet sent to recover the drone

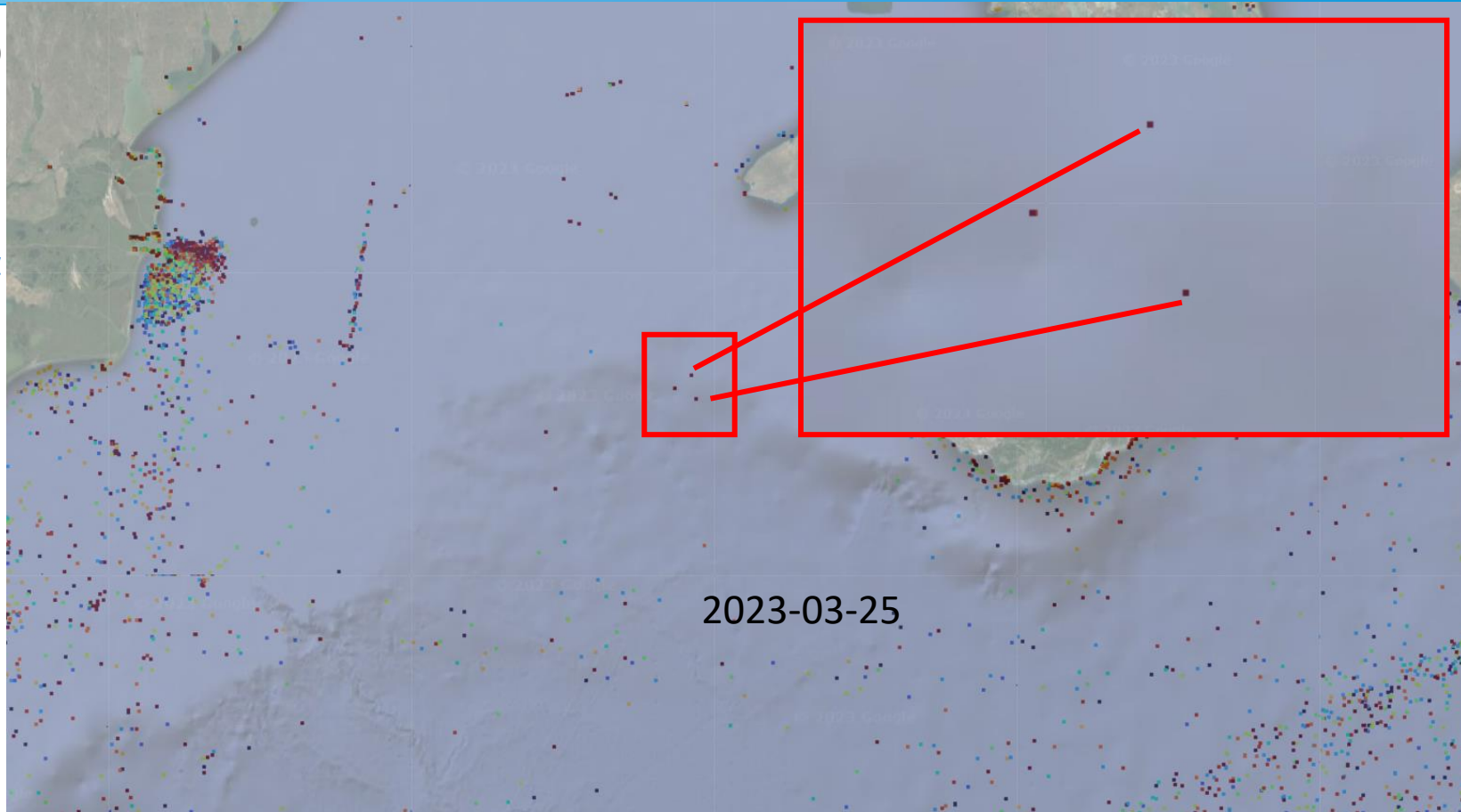


Applications – Naval Intelligence

- BBC reported two Russian ships sent to the crash site

(<https://www.bbc.com/news/world-us-canada-64972002>)

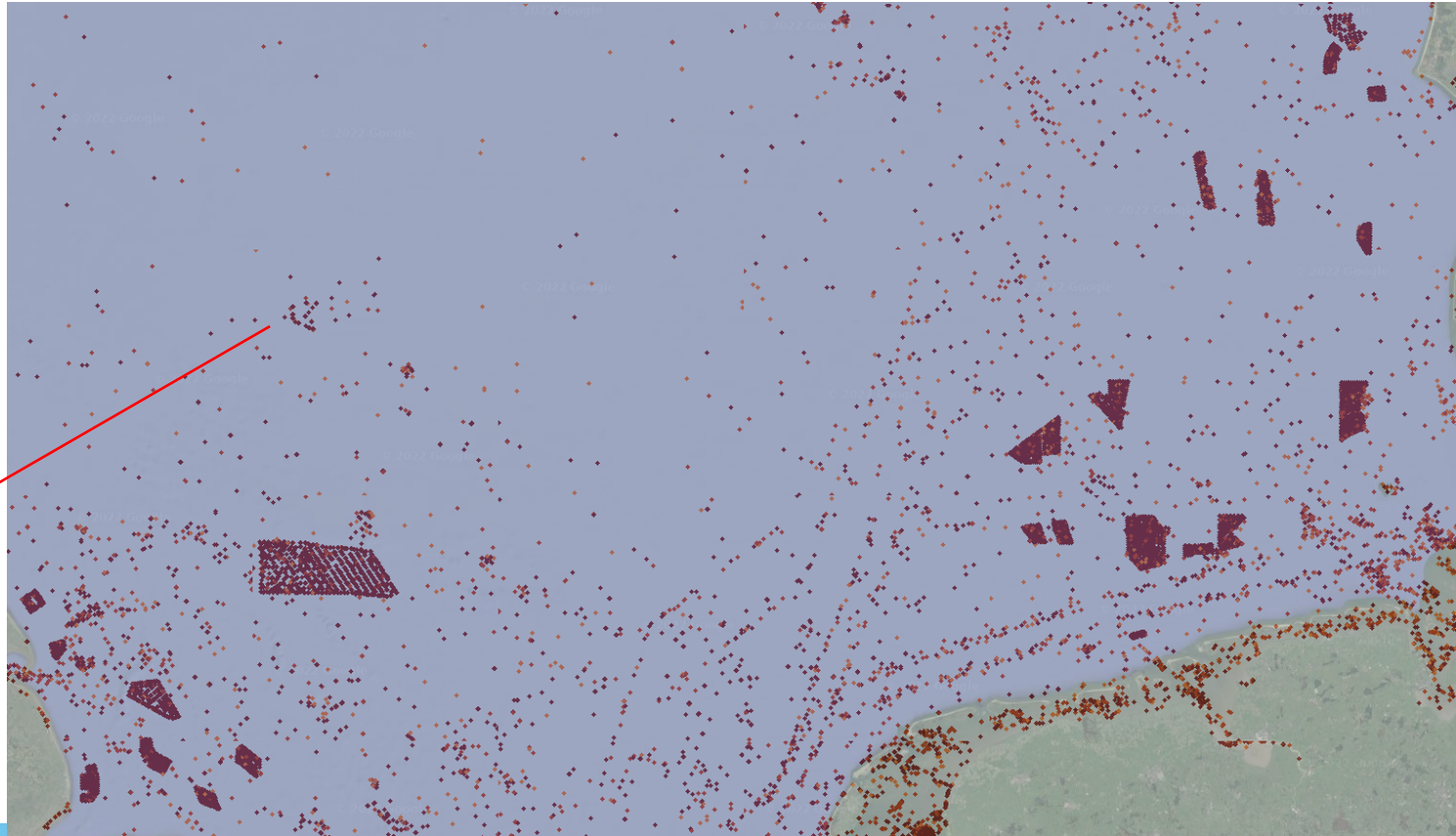
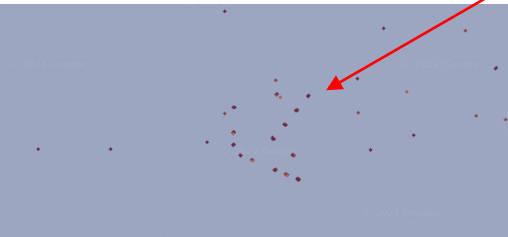
- These are likely those ships



Applications – Offshore wind power

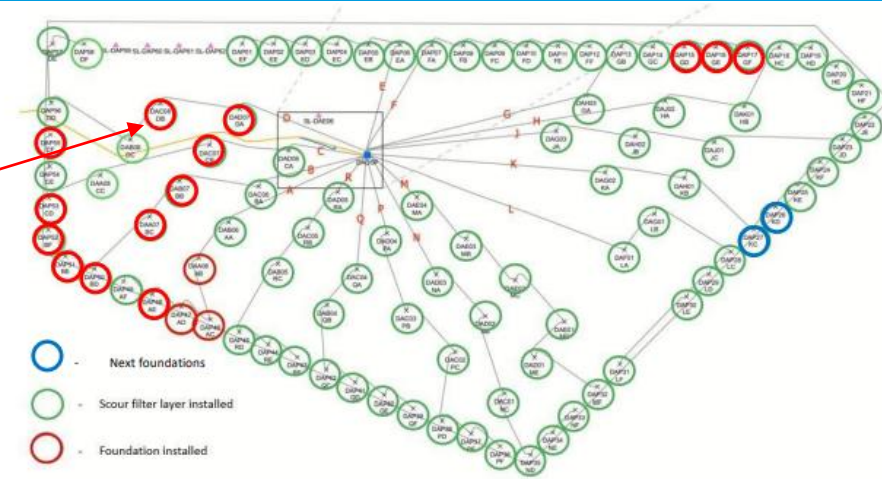
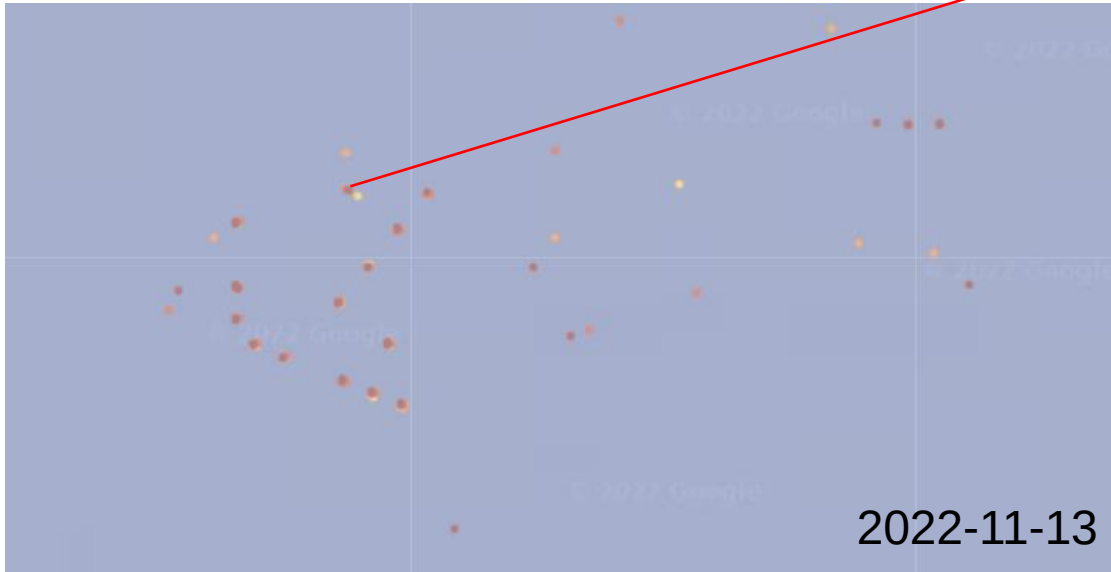
2022-11-13

- Wind farms appear as regular grids of objects detected that never move
- Monitoring wind farm construction
- Currently under construction
- Can see construction ships at the locations indicated in the notice to mariners issued



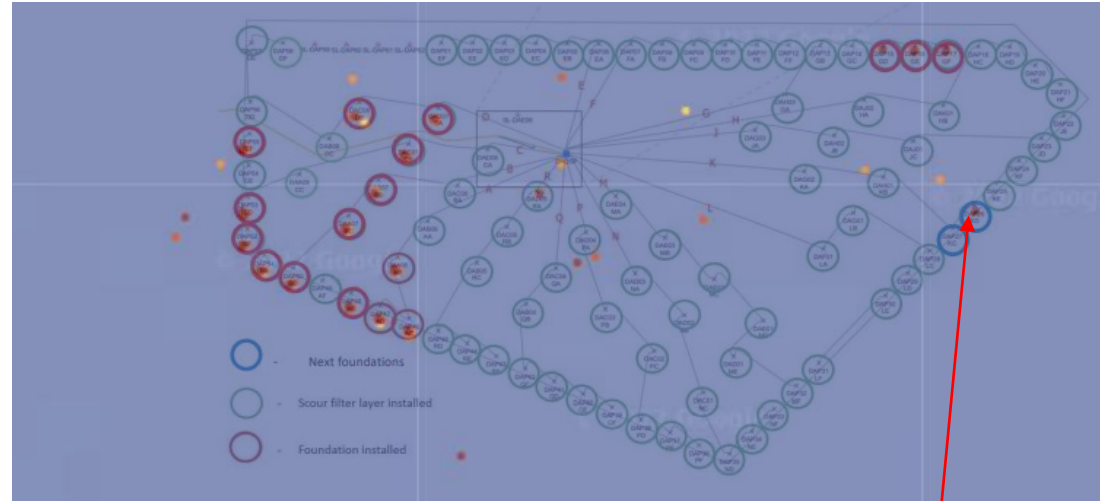
Applications – Offshore wind power

Monitoring wind farm construction



Applications – Offshore wind power

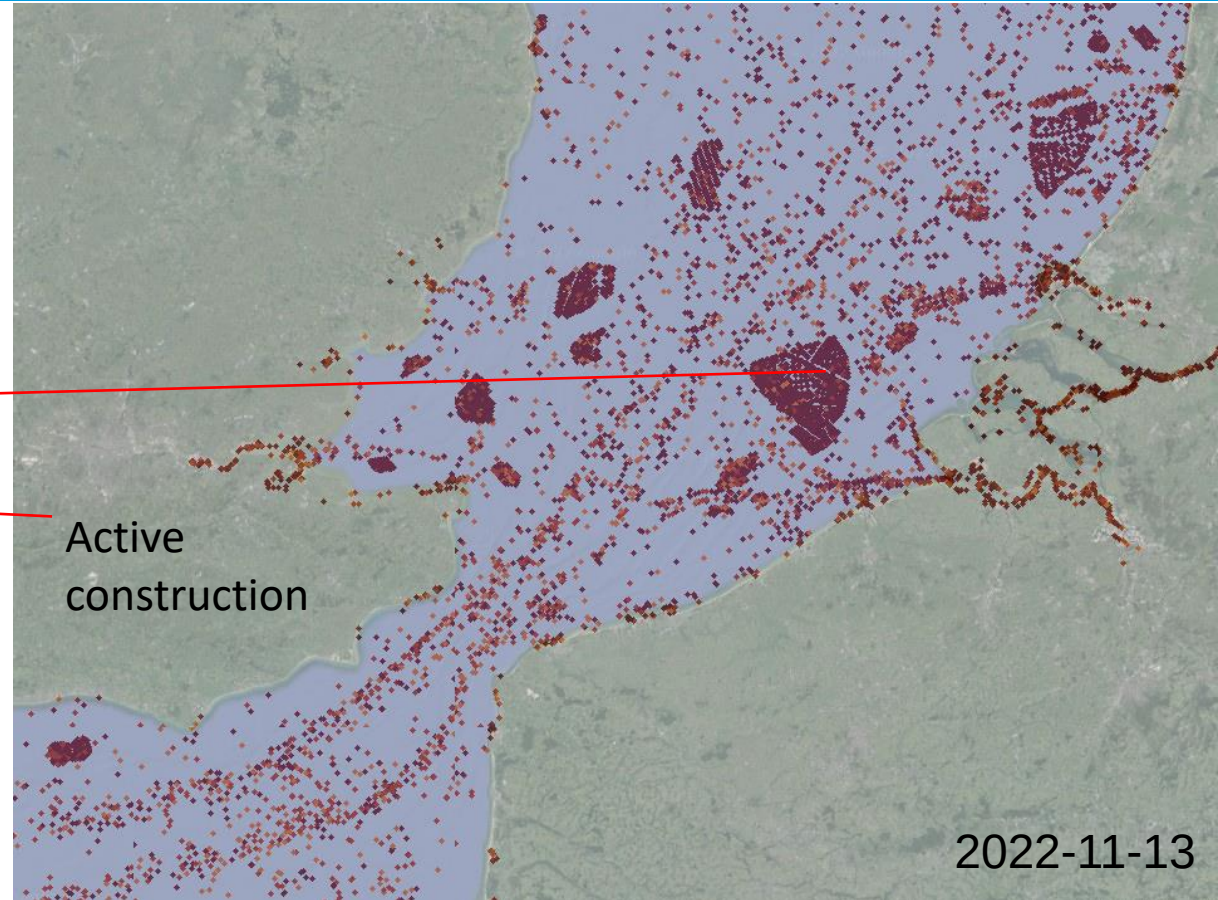
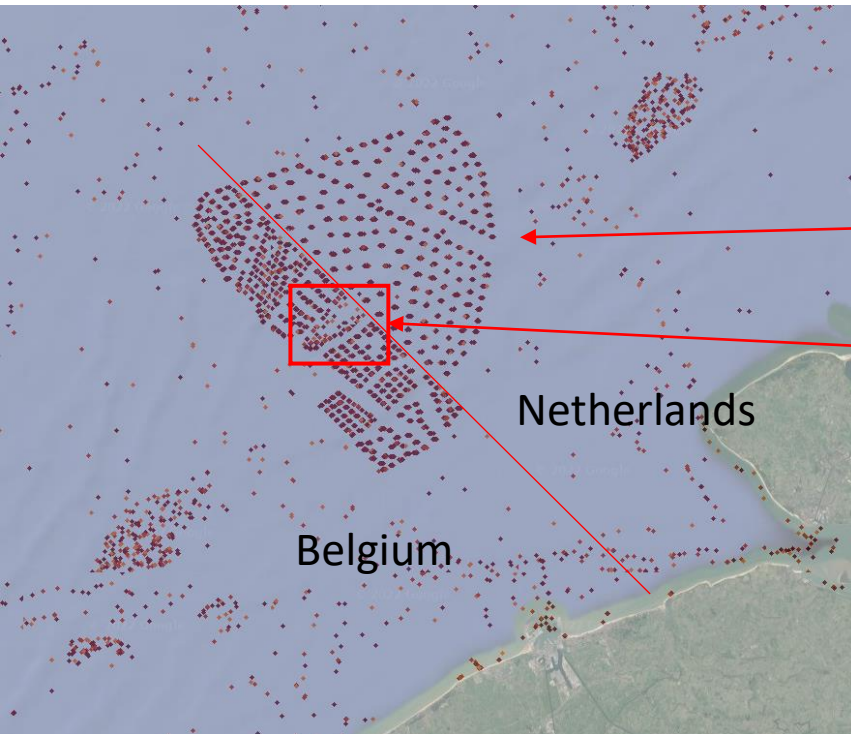
- Monitoring wind farm construction



Construction seems to have progressed from “Next” to “In Progress”

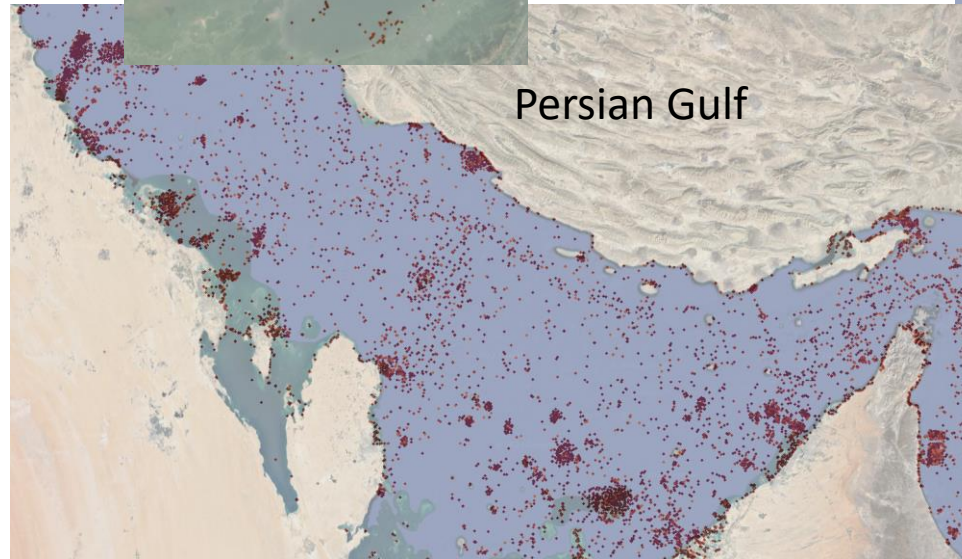
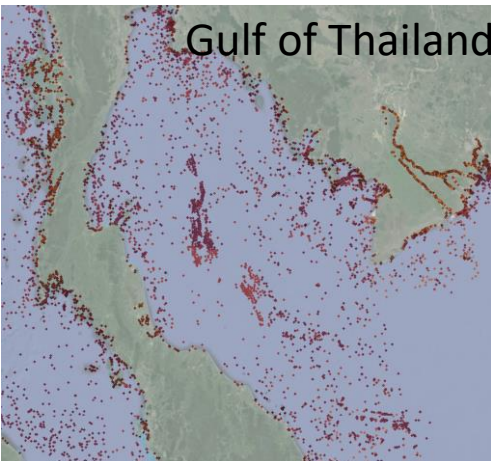
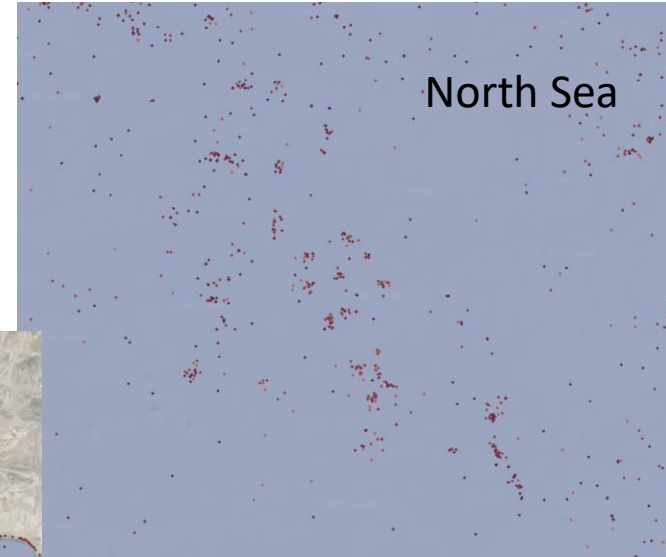
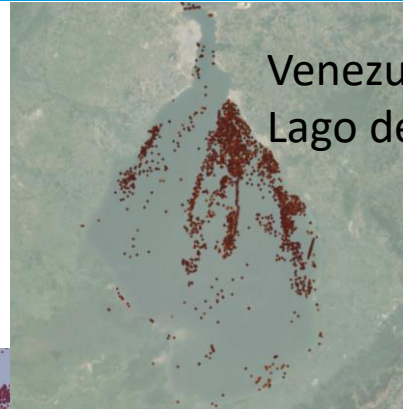
Applications – Offshore wind power

- Design differences between Belgian and Dutch wind projects



Applications – Offshore Oil Production

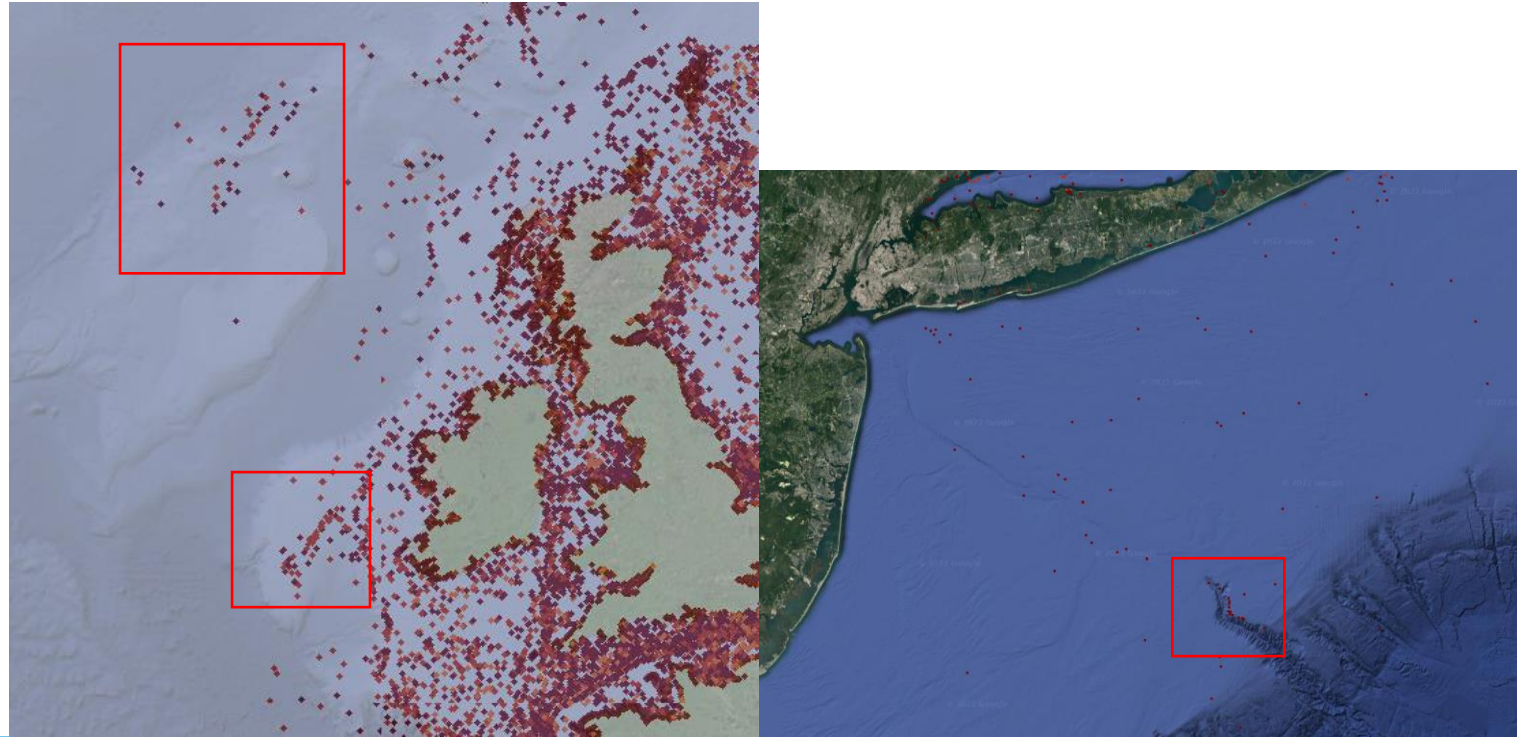
- Appear as irregular clusters of points around a static object
- Ships come and go around a static platform



2022-11-13

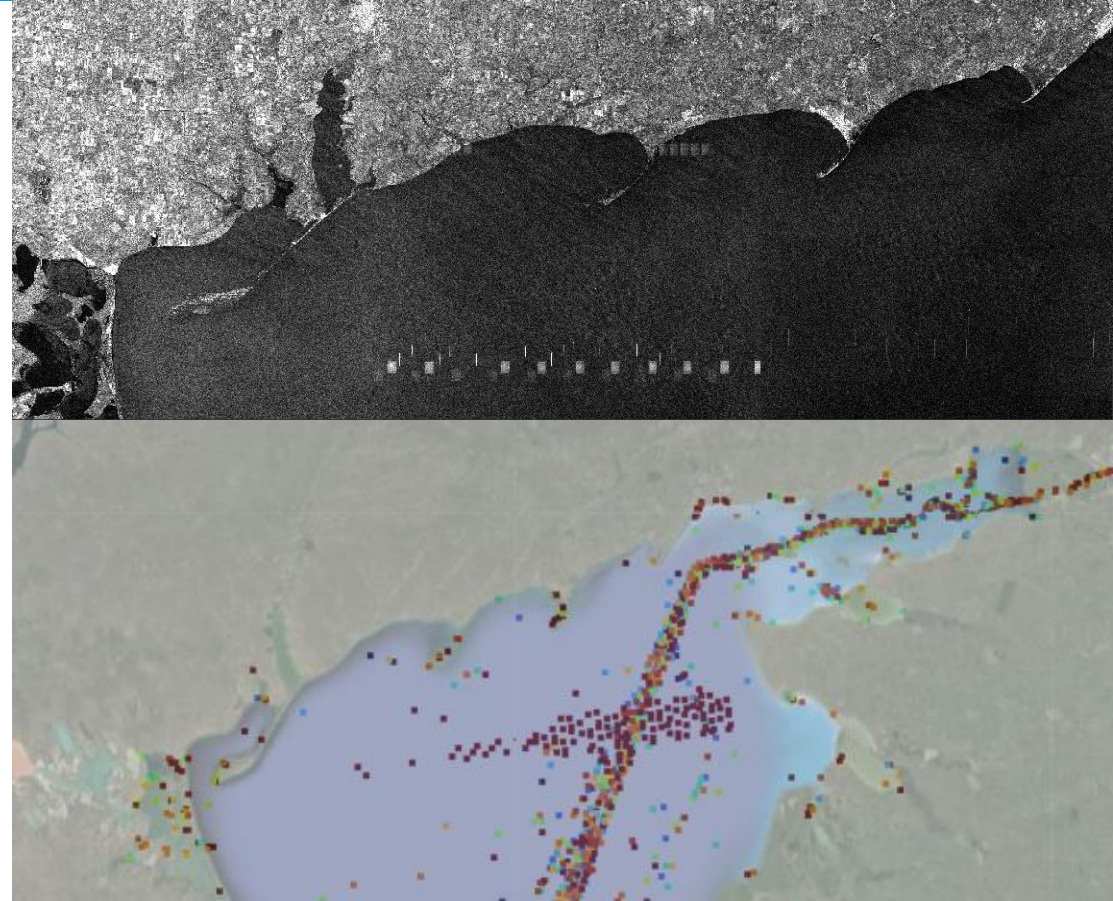
Applications – Fisheries

- Appear as clusters of ships, typically around a bank, always move between images



Applications – Active Radar Systems

- Active radar systems are known to cause interference in SAR imagery and since 2018 have been correlated with Air Defense systems according to a Bellingcat report.
- Appears in results as a distinct pattern of false detections



Next Steps

- Create simplified Sea Mask shapefiles so the download request can send a low vertex poly to query rather than a bounding box.
 - Will increase download efficiency so I don't download imagery over the middle of a continent or download imagery twice in overlapping bounding boxes.
 - This will allow me to include currently excluded AOIs in the South Indian and Pacific Oceans
- Include information on where SNT imagery is acquired
 - The user will see where no ships are located vs where no imagery is acquired.
- Include inland Seas and Large lakes (ie. Great Lakes and the Caspian Sea)
- Integrate other data sources to improve coverage and sampling
 - ie RCM, if possible to download by API

Acknowledgements

- Barry Carter, a Perl dev amongst men. Created the tiling script which makes the website usable. Without his help, it would have taken months more to get to where it is now.
- My friends and family who thought that this project was interesting enough to motivate me to not just give up when I ran into my first bug.
- My master's supervisors Dr. Braun and Dr. Fotopoulos.
- My colleagues at TRE Altamira who also thought this project was cool and gave me a platform to share it.
- OpenAI and ChatGPT who cut months out of the learning process on how to build a website.